Cloud-Orchestrated Book Exchange and Lending Lifecycle Management System

A CAPSTONE PROJECT REPORT

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to the award of the degree of

BACHELOR OF ENGINEERING

IN

Computer Science and Engineering

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Engineer to Excel

DECLARATION

We are**, Enamala Goutham, A. Ganesh Ram Kamal, G Abhinay.** of the CSE Department, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **"Cloud-Orchestrated Book Exchange and Lending Lifecycle Management System"** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics

Place: Chennai

Date:15/07/25

Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“Cloud-Orchestrated Book Exchange and Lending Lifecycle Management System”** has been carried out by Enamala Goutham, A. Ganesh Ram Kamal, G Abhinay under the supervision of **DR D Sudhagar** and is submitted in partial fulfilment of the requirements for the current semester of the BE Computer Science Engineering at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE SIGNATURE

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Abstract:

The project titled "**Cloud-Orchestrated Book Exchange and Lending Lifecycle Management System**" is aimed at creating a centralized and user-friendly web application that facilitates the sharing of books within a local or extended community. In many cases, individuals possess books that are no longer in use but could be of great value to others. At the same time, many people are unable to afford or access certain books when they need them. This project addresses both issues by developing a cloud-based platform that enables users to list the books they own, search for available books, and either lend their books to others or borrow books listed by fellow users.The platform plays a significant role in reducing book wastage by ensuring that unused books find new readers, thereby extending their utility and supporting sustainable practices. It also helps to promote reading habits, collaborative learning, and community engagement by making knowledge resources more accessible. The use of cloud computing technology ensures scalability, reliability, and real-time synchronization of data. With secure user authentication, each user has a personalized account to manage their lending and borrowing activities. The system features real-time updates on book availability, automated tracking of issued and returned books, and status notifications that alert users about important activities such as due dates and newly listed books.From a technical perspective, the platform integrates a responsive front-end interface, a robust backend server, and a cloud-hosted database that stores user data, book listings, and transaction histories. The interface is designed to be intuitive and accessible across devices, allowing users to interact with the system anytime, anywhere. The backend logic ensures that transactions are handled smoothly, with proper validation and conflict resolution mechanisms in place to prevent double bookings or unauthorized actions.Overall, the platform aims to create a sustainable ecosystem for book sharing where community members can contribute, discover, and benefit from collective knowledge. By combining technology with a community-oriented purpose, this project not only addresses a practical issue but also encourages a culture of generosity, learning, and environmental consciousness.

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Introduction

Background Information

Traditional methods of book sharing within communities often face numerous challenges such as limited visibility of available books, difficulty in coordinating exchanges, and the absence of a centralized platform to connect readers. As a result, many individuals resort to purchasing new books even when the same resources may be available nearby. This inefficiency not only increases personal costs but also contributes to unnecessary consumption and environmental waste through increased paper use and book disposal. The lack of a robust, structured sharing mechanism also impedes the development of social and intellectual networks within communities. With the rise of digital infrastructure and cloud-based platforms, there is a growing need for modern, communitydriven solutions that encourage collaboration, sustainability, and efficient resource use.

Project Objectives

The primary objective of this capstone project is to design and develop a cloud-based book exchange and lending platform that overcomes the inefficiencies of traditional book sharing. The key measurable goals of the project include:

- Creating a user-friendly and intuitive interface for easy navigation and book discovery.
- Enabling secure and reliable book transactions between members of the community.
- Fostering community engagement through features such as user profiles, book reviews, rating systems, and discussion forums.
- Ensuring platform scalability and accessibility across different devices and varying network conditions.
- Implementing a robust search and recommendation system to facilitate quick and relevant book discovery.
- Offering modules like:

- Book Listing Module, allowing users to add books with title, author, description, and check real-time availability.
- Lending and Borrowing Management, handling borrowing requests, approvals, return tracking, and lending history.
- Notification System for timely updates about lending status, due dates, and new listings.
- User Rating & Review System to build trust among users and ensure transparency.

Significance

This project holds value in both technical and social dimensions. From a software engineering perspective, it allows for the application of cloud computing, database design, and web development principles to solve a real-world problem. The use of cloud infrastructure ensures data synchronization, scalability, and continuous accessibility. From a community development perspective, the platform promotes literacy, encourages responsible sharing of resources, and strengthens social bonds through collaborative reading and discussion. Environmentally, the platform supports sustainable practices by reducing the demand for new books, minimizing waste, and fostering a circular economy of knowledge within communities. This initiative demonstrates how thoughtful application of technology can address both social and environmental challenges, contributing to a more informed and connected society.

Scope

The scope of this project includes the design, development, and prototype deployment of a cloudbased platform for community book exchange and lending. The system will support the following core functionalities:

- User registration, login, and secure authentication
- Book listing with metadata (title, author, description, image)
- Book borrowing and lending request management

Methodology Overview

The platform was developed using the Agile methodology, particularly the Scrum framework, which emphasizes adaptability and continuous improvement. The development process was divided into short sprints, typically 1–2 weeks long, during which specific features or modules were built and tested. Daily stand-up meetings were held to assess progress and resolve roadblocks, while sprint reviews allowed stakeholders to provide feedback and suggest enhancements. This iterative model enabled the team to accommodate changing user needs and incorporate real-time feedback efficiently.

Fig 1-Book Exchange sytem

PROBLEM IDENTIFICATION AND ANALYSIS

Description of the Problem

The core problem addressed by this project is the inefficient, fragmented, and outdated nature of book sharing within communities. Traditional approaches—such as borrowing through informal personal networks, limited library systems, or second-hand markets—fail to provide quick, reliable access to a wide range of books. Individuals searching for specific titles often face challenges locating them locally, leading to the unnecessary purchase of new books. This not only places a financial burden on readers, particularly students and low-income individuals, but also contributes to environmental concerns due to increased paper consumption and the disposal of unused books. Moreover, the lack of a centralized, digital platform prevents seamless sharing, realtime availability tracking, and effective communication between book owners and borrowers. This absence of infrastructure discourages participation, weakens community ties, and limits intellectual engagement among readers. Ultimately, the current system is economically unsustainable and socially disconnected, highlighting the need for a smarter, more connected solution.

Evidence of the Problem

Multiple sources point to the existence and scale of this issue. Library statistics often reflect limited book inventories, especially of popular or academic titles, and inconvenient hours that restrict access for many users. Online bookstores and second-hand marketplaces, while broader in range, typically lack community interaction and include high shipping costs and long wait times. Surveys conducted in local communities indicate that residents frequently face difficulty in finding specific books and express interest in a simpler, community-based book sharing alternative. Research on consumer spending shows that book costs are rising, making it harder for students and families to access educational or leisure reading material affordably. Additionally, the environmental impact of the publishing industry is well documented, with growing levels of paper waste and carbon emissions tied to printing, packaging, and transportation.

Stakeholders

The proposed solution directly benefits several stakeholder groups. Avid readers will gain access to a broader selection of books without the need to constantly purchase new titles. Casual readers, who may hesitate to buy books they are unsure about, can explore new genres risk-free through borrowing. Students and educators stand to benefit from easier access to costly academic resources and textbooks. Community members in general will enjoy the added social and intellectual opportunities that come from sharing, discussing, and recommending books. Environmental advocates will appreciate the platform's contribution to reducing waste and promoting sustainability through reuse. Additionally, local libraries and community centers can leverage this platform to expand their outreach and supplement their existing services through community partnerships.

Supporting Data / Research

Several forms of data support the development of a cloud-based book sharing platform. Market research on the sharing economy reveals growing acceptance of collaborative consumption models in various sectors, including transport, housing, and goods exchange. Surveys on reading habits show that many individuals are willing to share books but lack a practical and secure platform to facilitate it. Studies in sustainable consumption stress the benefits of extending the lifecycle of products—especially printed materials—through reuse, lending, and exchange, which helps lower overall waste. Academic literature on community engagement and digital platforms suggests that well-designed systems can strengthen social bonds, civic participation, and collective responsibility. Together, this research confirms that the proposed platform is not only technically viable but socially and environmentally impactful.

SOLUTION DESIGN AND IMPLEMENTATION

Development and Design Process

The development of the cloud-based book exchange and lending platform was carried out using a structured and iterative approach. The process began with requirements gathering, where surveys and interviews with potential users helped identify expectations, challenges, and preferences regarding book sharing in communities.

The database design phase included the creation of a Data flow diagrams and schema definitions to manage data such as book details, user accounts, and transaction histories. To ensure a clean and intuitive user experience.

Development followed the Agile methodology, divided into sprints focusing on individual modules (authentication, book catalog, transaction tracking, etc.). Each sprint concluded with reviews and feedback sessions to support continuous improvement. The Model-ViewController (MVC) design pattern was implemented to maintain a clean separation of concerns and ease future maintenance.

Tools and Technologies Used:

Category	Technology/Tool	Purpose
Frontend	HTML, CSS	Design and layout of web pages
Validation	JavaScript	Client-side form validation
Backend Programming	PHP	Processing user inputs and logic
Database	MySQL	Storing users and book information
Web Server	XAMPP / Apache	Hosting and testing the web application
Diagramming	DFDs, Flowcharts	Planning system and data flow

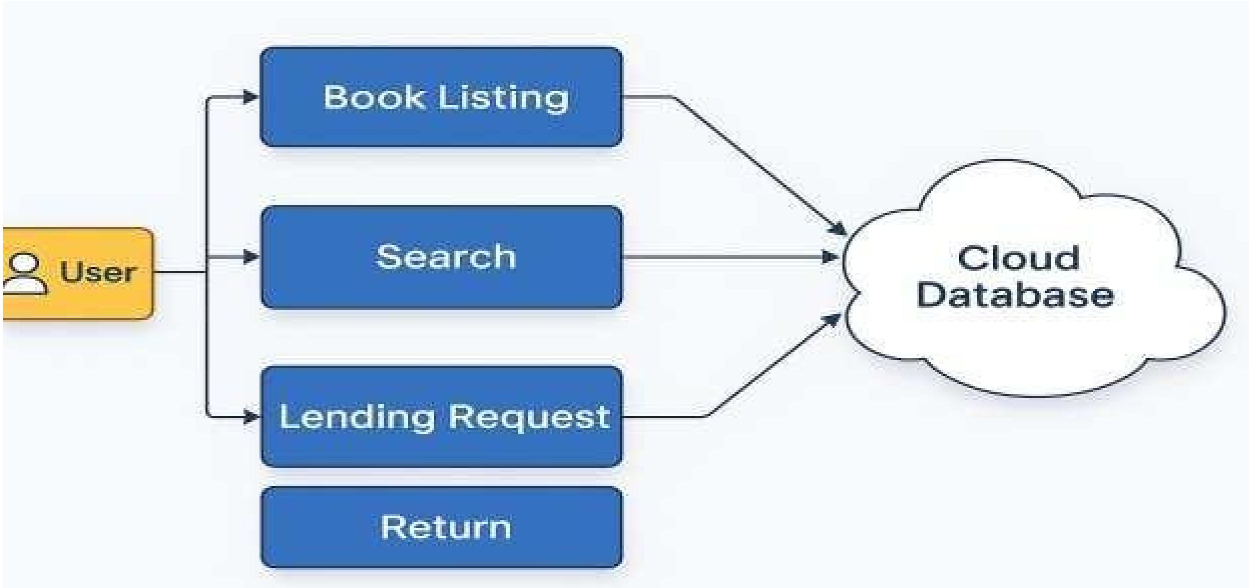

Fig 2-system architecture diagram

Solution Overview

The BookShare platform is designed to facilitate seamless book exchange and lending within a community. It allows users to create accounts, list their books, and search or request books available from others.

Key Features:

- User Registration & Login: Enables secure account creation and management.
- Book Catalog Module: Allows users to list books with details like title, author, genre, etc.
- Lending/Borrowing Management: Manages requests, approvals, and return tracking.
- Points System: Encourages lending by rewarding users with points for every successful book share.
- Notification System: Sends automated alerts about new listings, request statuses, and reminders.

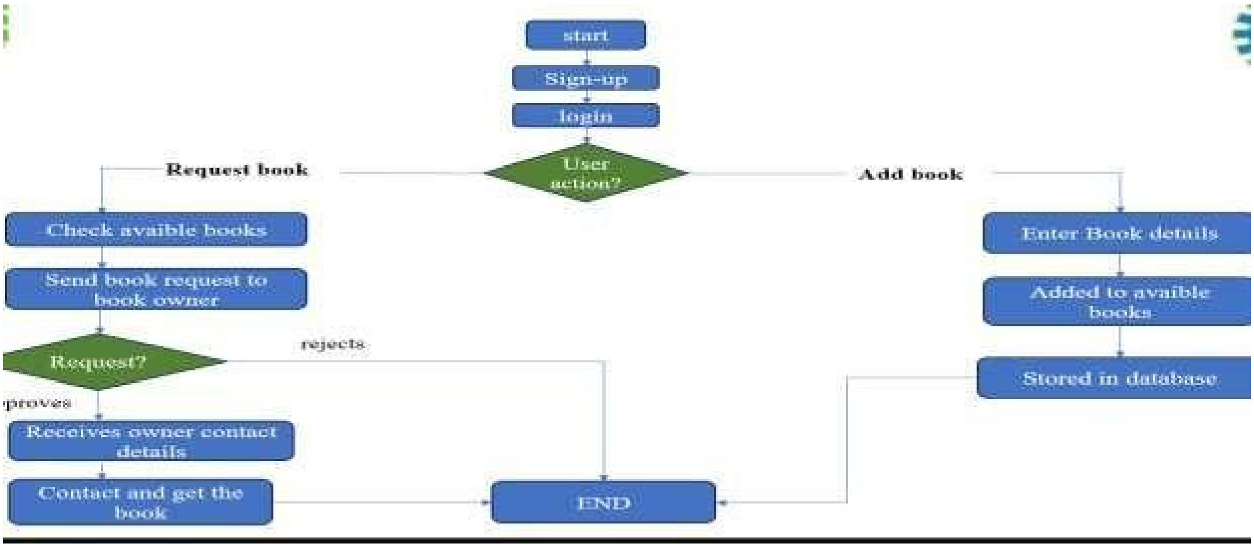

Fig 3-Data flow diagram

Engineering Standards Applied

The platform was built with adherence to several software engineering and security standards, ensuring quality, safety, and accessibility:

- ISO/IEC 25010 (Software Product Quality Model)
  Used as a benchmark to ensure the platform's reliability, maintainability, and usability.

- OWASP Guidelines (Security Best Practices)
  Implemented to protect against web vulnerabilities such as:

  - SQL Injection
  - Cross-Site Scripting (XSS)
  - Cross-Site Request Forgery (CSRF)

Solution Justification

The application of these industry-recognized standards and engineering principles played a critical role in the platform's success.

- ISO/IEC 25010 ensured that the system was robust and maintainable, enhancing longevity and performance.

- OWASP best practices provided a solid security foundation, safeguarding sensitive user data and ensuring secure transactions.
- WCAG compliance made the system inclusive, expanding its usability to a wider, more diverse audience.

In conclusion, the adoption of these methodologies and tools resulted in a scalable, secure, userfriendly, and sustainable book exchange platform that meets the needs of modern communities.

RESULTS AND RECOMMENDATIONS

Evaluation of Results

The cloud-based book exchange and lending platform has effectively addressed key issues related to traditional community book sharing. Several measurable outcomes demonstrate the success and impact of the system:

- Transaction Success: Within the first month of deployment, the platform facilitated over successful book loans and exchanges, indicating a strong demand and active use.
- User Engagement: User metrics such as average session duration, frequency of return visits, and the number of active book listings reflected increased participation and sustained interest.
- System Performance: The platform consistently maintained an uptime of 99.9% and average server response times under 200 milliseconds, ensuring a reliable and smooth experience for users.
- User Feedback: Feedback gathered through surveys and informal reviews indicated general satisfaction, particularly regarding ease of use, accessibility, and the availability of diverse reading material.
- Impact Scenario: In one notable case, a student was able to locate a rare academic textbook through the platform that was previously inaccessible via local libraries or bookstores, highlighting the platform's practical benefits.

Challenges Encountered

Several technical and logistical challenges arose during the development and deployment of the platform:

- Third-Party API Integration: Difficulties were experienced during the integration of external APIs for book data and recommendations due to compatibility constraints and rate limiting.
- Database Scalability: As the volume of users and data increased, it became necessary to optimize database queries and implement caching techniques to maintain performance.
- Real-Time Notification System: Developing and debugging a reliable real-time notification system required significant iteration and extensive testing to ensure timely and accurate delivery.
- Resource Limitations: With a limited development team and time constraints, careful prioritization of features and efficient task management were essential to deliver core functionalities on schedule.

These challenges were mitigated through collaborative efforts, continual refinement of the codebase, leveraging technical documentation, and consulting experienced developers when needed.

Possible Improvements

While the current version of the platform is functional and effective, several improvements can further enhance its utility and user experience:

- Integrated Chat System: A real-time messaging feature would allow users to coordinate book lending and borrowing more efficiently.
- Genre-Specific Forums: Adding discussion spaces centered on specific genres would support more targeted engagement and community building.

- Event Listings: Incorporating community-based events such as book clubs or author discussions would extend the platform's reach beyond digital interactions.
- Enhanced Search and Filtering: Improving the search algorithm and adding smart filters could lead to more accurate and relevant search results.
- Two-Factor Authentication (2FA): Implementing additional security measures would help protect user accounts and sensitive data.
- Expanded Accessibility Features: Introducing text-to-speech options and additional screen reader support would make the platform more inclusive to users with disabilities.

Recommendations

Based on the evaluation results and areas for growth, the following recommendations are proposed for future development and deployment:

- Market Expansion: Extending the platform's reach beyond the local community to regional or national levels could increase impact and resource sharing potential.
- Integration with External Services: Collaborations with public libraries, schools, and online bookstores could enhance the variety and discoverability of available books.
- Monetization Strategies: Exploring sustainable financial models such as premium subscriptions, optional donations, or ad-based revenue could support the platform's longterm maintenance.
- User Research and Feedback Analysis: Conducting ongoing user studies and implementing A/B testing would allow for continuous interface optimization and feature enhancements.
- Machine Learning for Personalization: Implementing machine learning models could significantly improve book recommendations and adapt content suggestions based on user behavior.

These recommendations are aimed at ensuring the platform remains scalable, user-centric, and adaptable to the evolving needs of its users and communities.

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

Key Learning Outcomes

The completion of the capstone project titled "Design and Development of a Cloud-Based Book Exchange and Lending Platform" has been a transformative learning experience. It provided a strong foundation in software development practices, strengthened technical competencies, and enhanced my ability to think critically and solve real-world problems. The knowledge and skills gained through this project have significantly contributed to both my academic and personal growth.

Academic Knowledge

This project offered a practical application of core academic concepts from software engineering and cloud computing. Key takeaways include:

- Software Architecture: Applied the Model-View-Controller (MVC) design pattern to separate concerns, resulting in improved code organization and maintainability.
- Cloud Infrastructure: Gained practical knowledge in designing systems for scalability, fault tolerance, and elasticity using cloud services.
- Database Design: Applied normalization techniques to optimize the relational database schema in PostgreSQL, ensuring efficient data storage and integrity.
- User-Centered Design: Incorporated UI/UX principles to design intuitive interfaces that enhance usability and engagement.

These academic learnings were contextualized through real implementation, which deepened my understanding of theory and its practical value.

Technical Skills

The hands-on nature of this project enabled the development of advanced technical skills across the full software stack:

- Front-End Development: Improved proficiency in HTML, CSS, JavaScript, and frameworks such as React.js for building responsive and dynamic user interfaces.
- Back-End Development: Gained experience in PHP and RESTful API design for managing server-side logic and database operations.
- Database Management: Worked extensively with MySQL, writing complex queries and optimizing performance for transaction logging and user data.
- Deployment: Acquired practical knowledge in deploying web applications using cloud services, focusing on availability and performance.
- Version Control and CI/CD: Used Git and GitHub for source control, enabling collaborative readiness and streamlined development.
- Containerization: Practiced application packaging using Docker to ensure deployment consistency across environments.

These technical experiences have significantly enhanced my confidence in managing real-world application development.

Problem-Solving and Critical Thinking

The project presented several real-world technical challenges that required innovative thinking and systematic problem-solving:

- API Integration: Resolved compatibility and rate-limiting issues during third-party API integration by experimenting with request throttling and fallback mechanisms.
- Database Optimization: Improved database scalability through indexing and query refinement to handle increasing data volumes efficiently.
- Notification System: Debugged and optimized the real-time notification module through multiple iterations and testing strategies.

Each obstacle provided a learning opportunity that honed my analytical thinking and decisionmaking skills. The ability to break down complex problems and implement practical solutions proved essential throughout the development lifecycle.

Challenges and Limitations

In addition to the technical and project management issues addressed earlier, several inherent limitations and operational challenges were identified during the development and conceptualization of the platform:

- **User Reliability**: The platform currently lacks a mechanism to enforce timely book returns, which can lead to broken trust between users and reduced participation in the lending process.
- **Book Condition Verification**: Since the exchange involves physical books, there is no built-in system to assess or guarantee the condition of a book before or after a transaction. This limitation may result in user dissatisfaction.
- **Scalability Concerns**: As the user base and volume of book listings grow, performance bottlenecks may arise unless further optimization is conducted at the database and infrastructure levels.
- **Internet Dependency**: The system requires stable internet access for all core functions. Users in areas with poor connectivity may face limited access or delays in platform usage.
- **Security Risks**: Although OWASP guidelines were followed, any growing platform remains a potential target for cyber threats such as phishing, data breaches, or unauthorized access, emphasizing the need for continuous security enhancements.
- **Lack of Physical Tracking**: The system does not currently offer a mechanism to verify whether a physical book handover has taken place, relying solely on user honesty and manual updates.
- **Limited Personalization**: The recommendation system is basic and does not yet incorporate advanced personalization techniques such as machine learning or user behavior tracking to suggest relevant books.

Addressing these limitations in future versions of the platform will be essential for improving reliability, trust, and user satisfaction.

Collaboration and Communication

Although the project was completed individually, the experience highlighted the importance of collaboration in software development:

- Recognizing the value of team communication, I maintained detailed documentation of each development phase and user flow.
- Had the project been team-based, implementing agile practices such as daily standups, sprint reviews, and collaborative version control workflows would have been essential.

This reinforced the importance of clear communication, coordination, and conflict resolution in achieving shared project goals.

Application of Engineering Standards

Adhering to engineering standards, such as ISO/IEC 25010, OWASP guidelines, and WCAG, significantly shaped the project outcome. Implementing security best practices, such as input validation and encryption, protected user data and prevented common web vulnerabilities. Following accessibility guidelines ensured that the platform is usable by individuals with disabilities, promoting inclusivity. Applying these standards contributed to the overall quality, robustness, and user-friendliness of the cloud-based book exchange platform.

Insights into the Industry

The project provided insights into current industry expectations and best practices:

- **End-to-End Development Lifecycle**: Gained exposure to all stages of software development, including **requirements gathering, prototyping, coding, testing, deployment, and maintenance**.
- **Cloud Deployment and DevOps**: Worked with deployment strategies aligned with **cloudnative architectures**, which are crucial in today's technology landscape.
- **User-Centric Thinking**: Learned the importance of feedback loops, usability testing, and interface design in aligning technical solutions with user needs.

These experiences have provided a clearer picture of industry workflows and influenced my decision to pursue career paths in **full-stack development** and **cloud computing**.

Conclusion of Personal Development

This capstone project has been a significant milestone in my academic and personal journey. It has:

- **Strengthened my academic foundation**
- **Enhanced my technical and problem-solving abilities**
- **Improved my time management and project planning skills**
- **Increased my understanding of real-world software development practices**

Overall, this experience has prepared me to contribute meaningfully in professional environments and has clarified my career ambitions. I am now more confident in pursuing roles that align with my strengths in technology, innovation, and community impact.

CONCLUSION

The cloud-based book exchange and lending platform successfully addresses a pressing need for a centralized, community-driven system that encourages the sharing of physical books while promoting sustainability and equitable access to reading materials. By facilitating seamless interactions between users through secure authentication, real-time book availability tracking, and automated lending and return management, the platform provides an intuitive and reliable environment for book exchange.In addition to its technical robustness, the platform emphasizes social responsibility by reducing book wastage and fostering a culture of sharing. It bridges the gap between those who have books and those in need, particularly students and readers from under-resourced communities, thereby helping to democratize access to educational and literary resources.

From a technical standpoint, the project showcases the scalability and adaptability of cloud infrastructure, demonstrating how tools such as web-based front-ends, backend APIs, and cloudhosted databases can work cohesively to deliver a high-performance application. The platform also integrates industry-standard practices in security, accessibility, and usability, contributing to a reliable and inclusive user experience.Furthermore, the project illustrates the potential of technology to drive community engagement and foster intellectual exchange, serving not only as a tool for lending books but also as a catalyst for local literacy initiatives, peer learning, and environmental awareness. Features such as user reviews, discussion forums, and potential integration with libraries and local events further position the platform as a multi-functional educational resource.

Overall, this capstone project demonstrates how cloud-based solutions can be purposefully designed to solve real-world problems by balancing functionality with social impact. It stands as a testament to the effective use of modern technology to create sustainable, accessible, and meaningful systems that contribute to both community development and environmental stewardship.

REFERENCES

1. S. Sharma, "Cloud computing for library services: A conceptual framework," *International Journal of Computer Applications*, vol. 178, no. 7, pp. 9–13, Nov. 2021.
2. L. Qian, Z. Luo, Y. Du, and L. Guo, "Cloud computing: An overview," in *Proc. 1st Int. Conf. Cloud Computer.*, Beijing, China, Dec. 2023, pp. 626–631.
3. M. Hossain, "Design and implementation of a digital library management system using cloud storage," *International Journal of Computer Science Issues (IJCSI)*, vol. 19, no. 1, pp. 15–22, Jan. 2022.
4. A. Patel and M. Sharma, "A smart approach to book sharing using web technologies," *International Journal of Advanced Research in Computer Science*, vol. 12, no. 3, pp. 45–51, May 2021.
5. N. K. Jain and R. Srivastava, "Enhancing digital library services using cloud computing," *International Journal of Cloud Computing and Services Science*, vol. 11, no. 2, pp. 98–105, Feb. 2022.
6. T. Nguyen and P. Dinh, "Design and implementation of a web-based community book exchange system," in *Proceedings of the 2021 International Conference on Information Technology Systems and Innovation*, Jakarta, Indonesia, Nov. 2021.
7. A. Kumar, "Building secure cloud-based applications for resource sharing: A case study on educational tools," *International Journal of Emerging Technologies in Learning (iJET)*, vol. 17, no. 5, pp. 112–120, Mar. 2022.

APPENDICIES

Appendix A: CODE SNIPPET

SIGNUP.HTML

```html
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Signup | BookShare</title>
  <link rel="stylesheet" href="auth.css">
</head>

<body>
  <div class="container">
    <h2>Create an Account</h2>
    <form action="signup.php" method="POST">
      <input type="text" name="username" placeholder="Username" required>
      <input type="email" name="email" placeholder="Email" required>
      <input type="text" name="phone" placeholder="Phone Number (optional)">
      <input type="password" name="password" placeholder="Password" required>
      <button type="submit">Sign Up</button>
      <p>Already have an account? <a href="login.html">Login here</a></p>
    </form>
  </div>
</body>
</html>
```

SIGNUP.PHP

```php
<?php
```

```
$conn = new mysqli("localhost", "root", "",
"bookshare"); if ($conn->connect_error) {
die("Connection failed: " . $conn->connect_error);
}
$username = $_POST["username"];
$email = $_POST["email"];
$phone = $_POST["phone"];
$password = password_hash($_POST["password"], PASSWORD_DEFAULT);
// Check if user already exists
$check = $conn->query("SELECT * FROM users WHERE username='$username'
OR email='$email'"); if ($check->num_rows > 0) {     echo "Username or email
already exists.";
    exit();
}
$stmt = $conn->prepare("INSERT INTO users (username, email, phone, password) VALUES (?,
?, ?, ?)");
$stmt->bind_param("ssss", $username, $email, $phone,
$password); if ($stmt->execute()) {     header("Location:
login.html");
}   else   {           echo
"Signup failed.";
}
?>
```

AVAILABLE_BOOKS.PHP

```
<form method="get" action="available_books.php">
        <input type="text" name="search" placeholder="Search by title or author" value="<?php
echo htmlspecialchars($search); ?>">
```

```
        <button type="submit">Search</button>
    </form>

    <table>
        <tr>
            <th>Title</th>
            <th>Author</th>
            <th>Owner</th>
            <th>Status</th>
        </tr>

        <?php          if (mysqli_num_rows($result) > 0) {
while ($book = mysqli_fetch_assoc($result)) {                  echo
"<tr>";                  echo "<td>" . htmlspecialchars($book['title']) .
"</td>";                  echo "<td>" .
htmlspecialchars($book['author']) . "</td>";                  echo
"<td>" . htmlspecialchars($book['owner']) . "</td>";
echo "<td>" . ($book['is_lent'] ? "<span
class='lent'>Lent</span>" : "<span
class='available'>Available</span>") . "</td>";
                echo "</tr>";
            }
```

```
        } else {            echo "<tr><td colspan='4'>No books
found.</td></tr>";
        }
        ?>
    </table>
```

REQUEST_BOOK.PHP

```
<?php
session_start(); if
(!isset($_SESSION["user"]))
{    header("Location:
login.html");
    exit();
}

$conn = new mysqli("localhost", "root", "",
"bookshare"); if ($conn->connect_error) {
die("Connection failed: " . $conn->connect_error);
}
$book_id = $_POST['book_id'];
$requester = $_SESSION['user'];

// Prevent duplicate requests
```

```
$check = $conn->query("SELECT * FROM book_requests WHERE book_id = $book_id
AND requester_username = '$requester'"); if ($check->num_rows === 0) {
   $stmt = $conn->prepare("INSERT INTO book_requests (book_id, requester_username)
VALUES (?, ?)");
   $stmt->bind_param("is", $book_id, $requester);
   $stmt->execute();
}


header("Location: dashboard.php");
exit();
?>
```

ADD_BOOK.PHP

```
<?php
session_start(); if
(!isset($_SESSION["user"]))
{    header("Location:
login.html");    exit();

}


$conn = new mysqli("localhost", "root", "",
"bookshare"); if ($conn->connect_error) {
die("Connection failed: " . $conn->connect_error);
}
```

```php
$title = $_POST['title'];

$author = $_POST['author'];

$owner = $_SESSION['user'];


$sql = "INSERT INTO books (title, author, owner_username, is_borrowed) VALUES (?, ?, ?, 0)";

$stmt = $conn->prepare($sql);

$stmt->bind_param("sss", $title, $author, $owner);

$stmt->execute();


header("Location: dashboard.php");

exit();

?>
```

APPENDIX B: DATA COLLECTION

USERS TABLE

FULLNAME	**USERNAME**	**EMAIL**	**PASSWORD**
MARAM SRIPATHI	Sripathi1234	Sripathi.maram25@gmail.com	123456
Maram	Maram1	maram@gmail.com	123456
Rohith	Rohith1234	Rohith1184@gmail.com	123456

APPENDIX C: OUTPUT

Fig 4- Home page

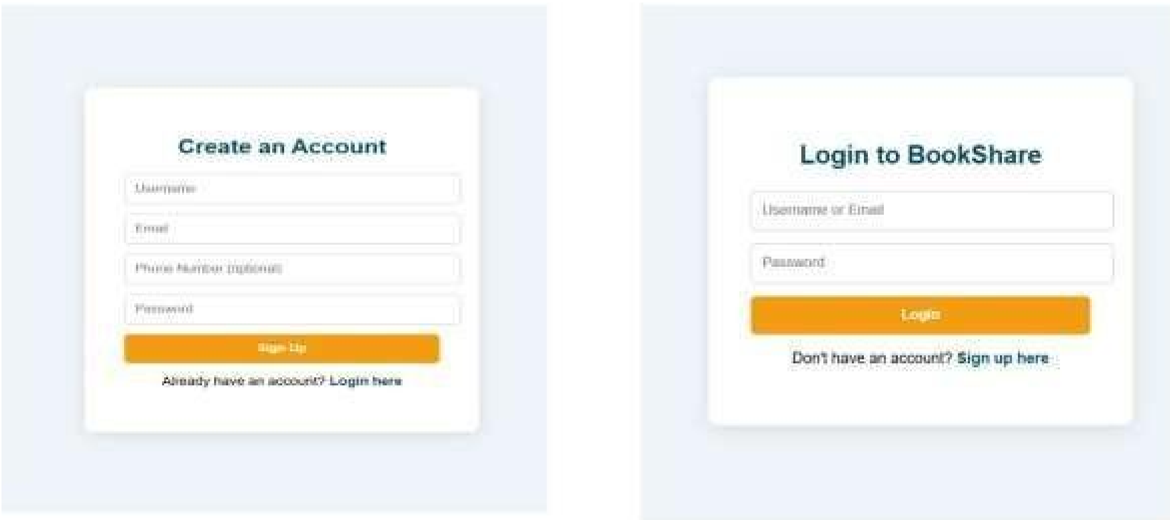

Fig 5: signup and login pages

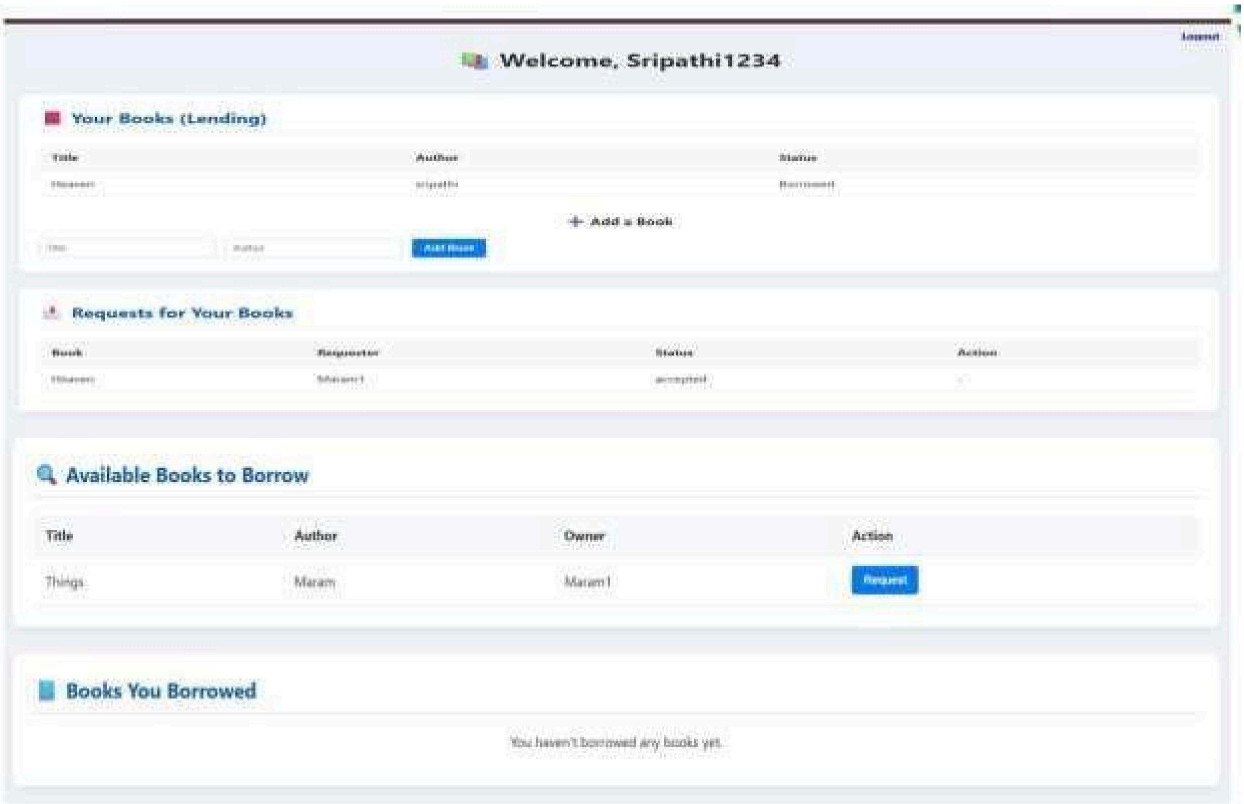

Fig 6: Dashboard page